A PERSONALIZED REAL-TIME PROACTIVE VOICE MEMORY ASSISTANT

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Motivation

Timely recall of details

- is crucial for collaboration, decision-making, and planning;
- can potentially benefit people with dementia (57 million globally) or memory impairments (10% of adults over 65 have dementia and 22% have mild impairment in U.S.).

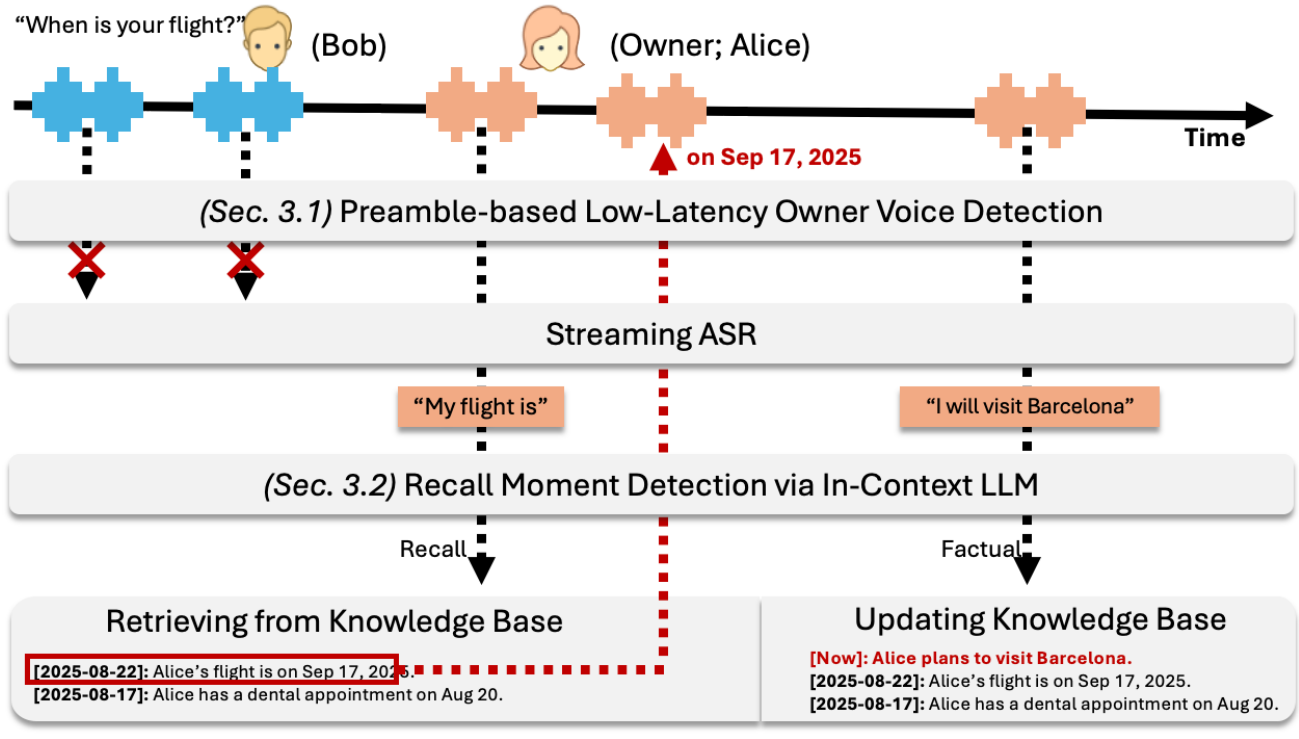
Research Gap

System	Proactive Assistant	Owner-awareness Privacy	In-conversation Summarization
Memoro [12]	✔	✖	✖
MemPal [5]	✖	✖	✖
Mirai [13]	✔	✖	✖
LLAMAPIE [6]	✔	✖	✖
MemoryAids (Ours)	✔	✔	✔

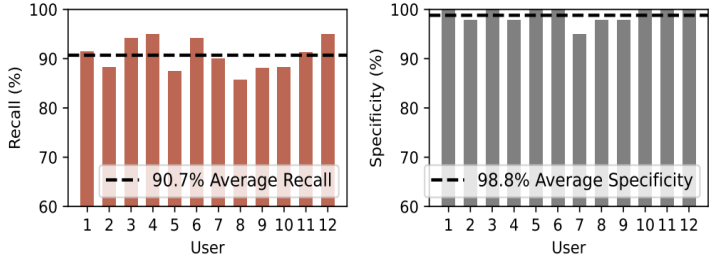
Controlled Study

- N = 12
- Predefined Dialogue
- containing numerous factual details, such as flight timings, locations, etc.
- Real-time memory assistant

Our Solutions



Evaluation on Accuracy:



Evaluation on Latency (ms):

Component	Mean ± Std	Median	95%-tile
Owner Detection	53.2 ± 14.7	51.0	78.7
Transcription (ASR)	79.8 ± 20.2	80.2	112.9
RAG (Embed+LLM)	793.9 ± 82.6	793.6	929.5
Total	926.9 ± 87.7	926.4	1070.2

Evaluation on User-Satisfaction:

